

OKHANSK, Russian Federation

WMO: 283210

Lat: 57.717N

Lon: 55.383E

Elev: 163

StdP: 99.38

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -26.5	(d) -32.4	(e) 0.2	(f) -29.3	(g) -29.3	(h) 0.3	(i) -26.2	(j) 7.1	(k) -7.3	(l) 6.4	(m) -8.9	(n) 1.6	(o) 180	(p) 0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	29.7	20.4	27.9	19.4	26.0	18.4	21.7	27.4	20.6	26.0	19.5	24.4	1.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.8	24.1	18.7	13.8	23.1	17.6	12.9	22.1	64.0	27.3	60.0	26.0	56.3	24.3	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.5	(b) 5.7	(c) 5.1	(e) -33.8	(f) 32.4	(g) 3.6	(h) 2.2	(i) -36.3	(j) 34.0	(k) -38.4	(l) 35.3	(m) -40.4	(n) 36.6	(o) -43.0	(p) 38.2		
(a) 6.5	(b) 5.7	(c) 5.1	(e) -33.9	(f) 23.3	(g) 3.5	(h) 1.6	(i) -36.4	(j) 24.4	(k) -38.5	(l) 25.3	(m) -40.5	(n) 26.2	(o) -43.0	(p) 27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-13.2	-11.8	-4.3	4.0	11.7	16.7	19.2	16.3	10.5	3.4	-4.8	-10.8		
	DBStd	12.61	7.36	6.96	4.67	4.90	5.21	4.33	4.12	4.12	4.31	4.55	6.32	7.45		
	HDD10.0	3358	721	611	443	189	44	4	0	2	44	210	445	645		
	HDD18.3	5676	979	844	701	430	213	81	41	91	236	462	695	903		
	CDD10.0	859	0	0	0	9	96	205	286	197	60	6	0	0		
	CDD18.3	135	0	0	0	0	7	32	68	27	2	0	0	0		
	CDH23.3	1131	0	0	0	1	81	268	577	194	11	0	0	0		
	CDH26.7	295	0	0	0	0	13	56	172	53	1	0	0	0		
(14) Wind	WSAvg	2.2	2.3	2.3	2.5	2.2	2.3	2.0	1.7	1.9	2.0	2.5	2.4	2.4		
(15) Precipitation	PrecAvg	592	40	28	32	32	48	70	66	69	60	56	47	42		
	PrecMax	775	84	50	54	58	106	128	189	150	124	97	91	75		
	PrecMin	411	9	12	2	1	15	33	5	8	10	21	10	24		
	PrecStd	77	16	11	15	17	23	26	37	30	28	21	20	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	1.4	8.2	22.4	28.4	30.3	32.3	31.4	24.9	17.7	8.2	2.0		
		MCWB	0.4	0.1	3.4	12.9	16.5	20.3	21.2	20.4	16.0	11.3	7.2	1.4		
	2%	DB	-0.2	0.4	5.0	17.9	25.6	28.2	30.2	27.6	21.4	14.4	5.8	0.9		
		MCWB	-0.6	-0.4	1.5	10.2	15.2	19.1	20.9	19.7	14.9	10.1	5.0	0.3		
	5%	DB	-1.5	-0.6	3.1	14.4	23.1	26.5	28.6	25.2	19.1	11.7	3.5	0.0		
		MCWB	-2.0	-1.2	0.5	8.2	14.1	18.6	20.3	18.8	14.2	9.2	2.8	-0.4		
	10%	DB	-3.4	-2.2	1.6	11.5	20.6	24.5	26.8	23.0	16.7	9.6	1.8	-1.6		
		MCWB	-3.9	-3.0	-0.1	6.6	12.8	17.3	19.3	17.8	12.9	7.9	1.1	-2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.6	4.1	13.1	18.2	22.0	24.5	22.2	17.4	12.4	7.4	1.7		
		MCDB	0.8	0.9	7.2	21.0	25.6	27.9	29.3	27.4	22.0	15.5	8.1	1.9		
	2%	WB	-0.6	-0.2	2.2	10.8	16.4	20.6	22.4	20.9	15.8	10.6	5.0	0.5		
		MCDB	-0.2	0.3	4.0	16.6	23.7	26.6	28.1	26.5	19.7	13.6	5.6	0.8		
	5%	WB	-1.9	-1.2	1.1	8.8	15.0	19.3	21.1	19.6	14.7	9.3	2.9	-0.5		
		MCDB	-1.4	-0.6	2.5	14.0	21.3	24.6	26.6	23.9	17.8	11.3	3.3	0.0		
	10%	WB	-3.8	-3.0	0.2	7.0	13.6	18.1	20.1	18.4	13.6	8.0	1.2	-2.1		
		MCDB	-3.4	-2.2	1.5	11.0	19.1	23.2	25.1	21.9	16.3	9.7	1.6	-1.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	6.8	7.4	8.7	10.3	9.4	9.4	8.1	6.9	4.7	4.0	5.0		
		MCDBR	5.1	5.2	8.1	13.3	14.0	12.2	12.1	11.3	10.3	8.4	4.3	4.0		
	5% WB	MCWBR	5.1	4.8	5.7	7.2	6.4	5.4	4.8	5.0	5.1	4.8	3.9	4.1		
		MCWBR	5.1	5.0	6.7	12.1	12.2	11.1	10.7	10.1	8.7	7.6	3.9	4.0		
(40) Clear-Sky Solar Irradiance	taub		0.291	0.317	0.372	0.401	0.391	0.388	0.409	0.425	0.377	0.348	0.308	0.284		
	taud		2.001	2.009	1.929	2.069	2.299	2.361	2.310	2.237	2.340	2.272	2.100	1.971		
	Ebn at Noon		512	683	742	798	843	851	820	772	758	661	495	390		
	Edh at Noon		60	94	135	138	118	113	116	117	91	73	54	46		
(44) All-Sky Solar Radiation	RadAvg		0.55	1.45	2.78	4.12	5.18	5.52	5.54	4.15	2.46	1.06	0.44	0.30		
	RadStd		0.09	0.23	0.33	0.52	0.57	0.49	0.60	0.37	0.29	0.16	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	+1.06	+1.08	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page